Cadmium in organic and conventional pig production.

The main sources of cadmium (Cd) input to soils have been phosphate fertilizers and deposition from air. In organic farming, phosphate fertilizers are not used, which may in the long term result in lower Cd levels. In the present study, feed, kidney, liver, and manure from growing/finishing pigs raised conventionally and organically on the same farm were microwave-digested and analyzed for Cd by graphite furnace atomic absorption spectrometry. Cd was also analyzed in soil and water. A quality control program was included. The organic pigs (n = 40) were raised outdoors and fed an organic feed; the conventional pigs (n = 40) were raised indoors and given a conventional feed. The Cd levels in organic and conventional feed were 39.9 microg/kg and 51.8 microg/kg, respectively. Organic feed contained 2% potato protein, which contributed 17% of the Cd content. Conventional feed contained 5% beet fiber, which contributed 38% of total Cd content. Both feeds contained vitamin-mineral mixtures with high levels of Cd: 991 microg/kg in organic and 589 microg/kg in conventional feed. There was a significant negative linear relationship between Cd concentration in kidney and kidney weight. There was no significant difference in liver Cd levels between organic and conventional pigs and the mean +/- SD was 15.4 +/- 3.0. In spite of the lower level of Cd in the organic feed, the organic pigs had significantly higher levels in kidneys than the conventional pigs, 96.1 +/- 19.5 microg/kg wet weight (mean +/- SD; n = 37) and 84.0 +/- 17.6 microg/kg wet weight (n = 40), respectively. Organic pigs had higher Cd levels in manure, indicating a higher Cd exposure from the environment, such as ingestion of soil. Differences in feed compositions and bioavailability of Cd from the feed components may also explain the different kidney levels of Cd.